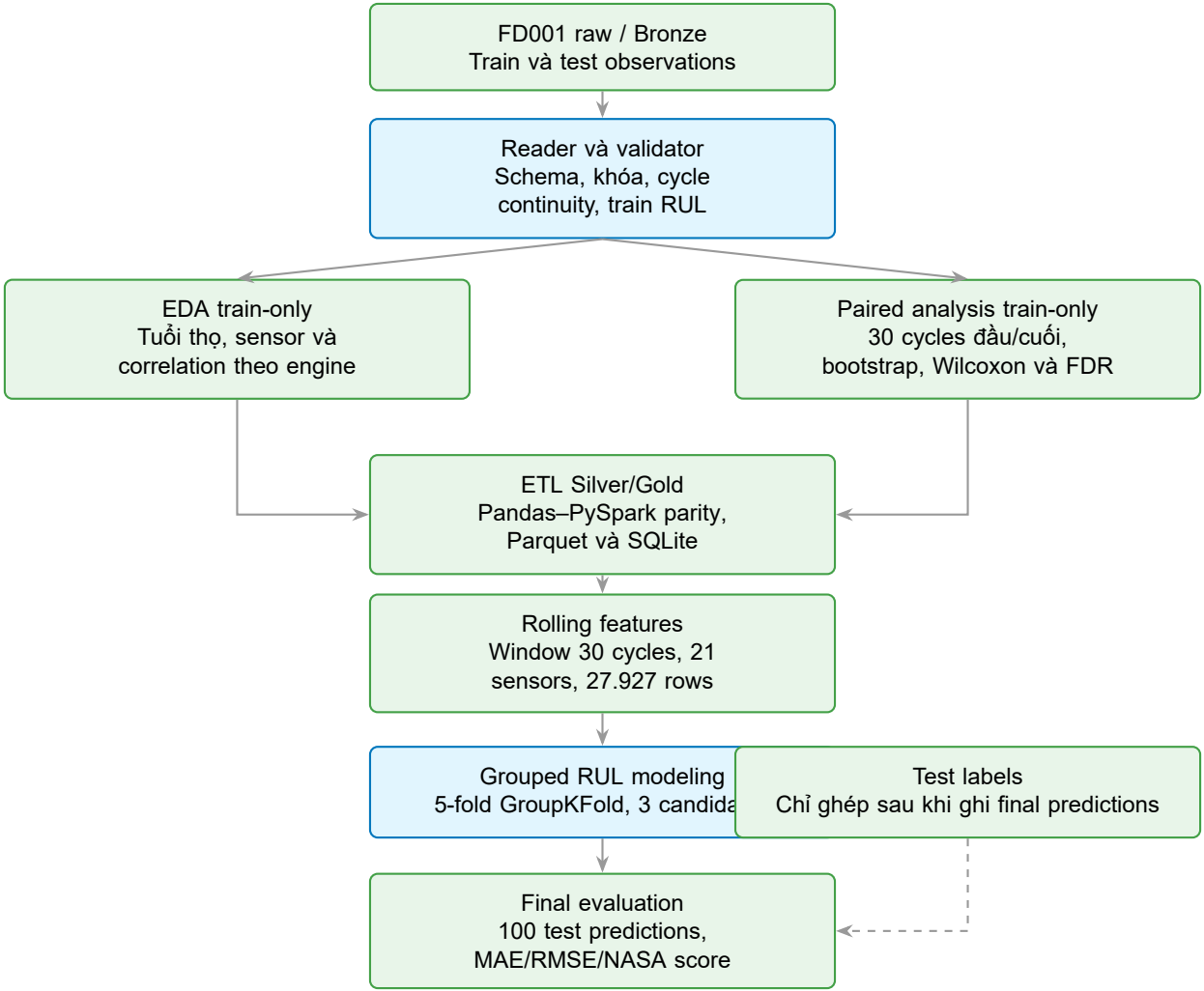
FD001 raw / Bronze
Train và test observations
Reader và validator
Schema, khóa, cycle
continuity, train RUL
EDA train-only
Tuổi thọ, sensor và
correlation theo engine
Paired analysis train-only
30 cycles đầu/cuối,
bootstrap, Wilcoxon và FDR
ETL Silver/Gold
Pandas–PySpark parity,
Parquet và SQLite
Rolling features
Window 30 cycles, 21
sensors, 27.927 rows
Grouped RUL modeling
5-fold GroupKFold, 3 candida
Test labels
Chỉ ghép sau khi ghi final predictions
Final evaluation
100 test predictions,
MAE/RMSE/NASA score